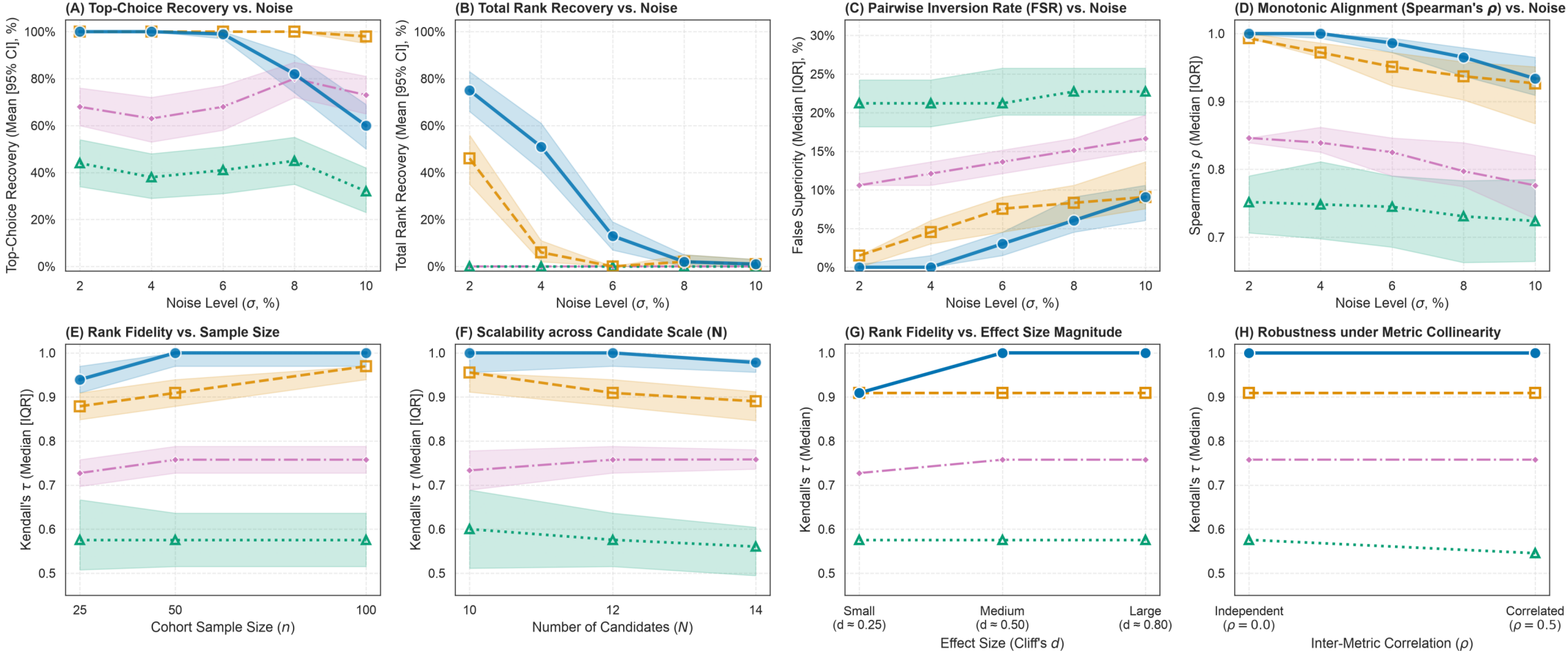


Benchmark Validation Overview

Default parameters (unless otherwise specified):  $N = 12$ ,  $n = 50$ , Cliff's  $d \approx 0.50$ ,  $\rho = 0.0$ ,  $K = 100$  runs per point

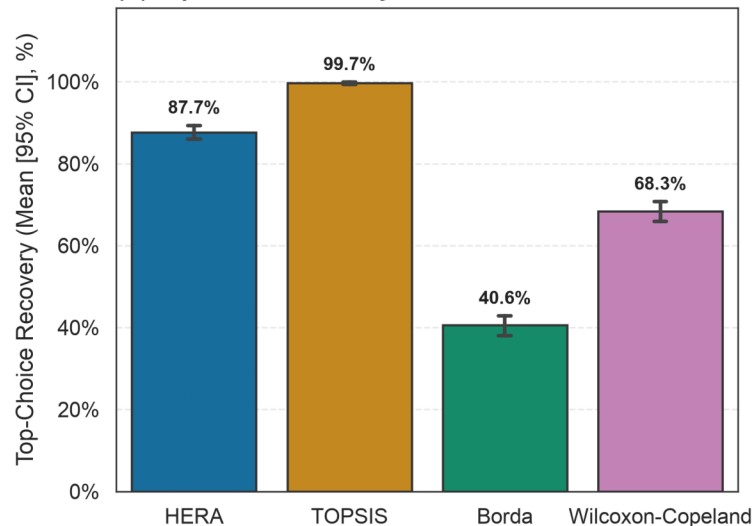


HERA TOPSIS Borda Wilcoxon-Copeland

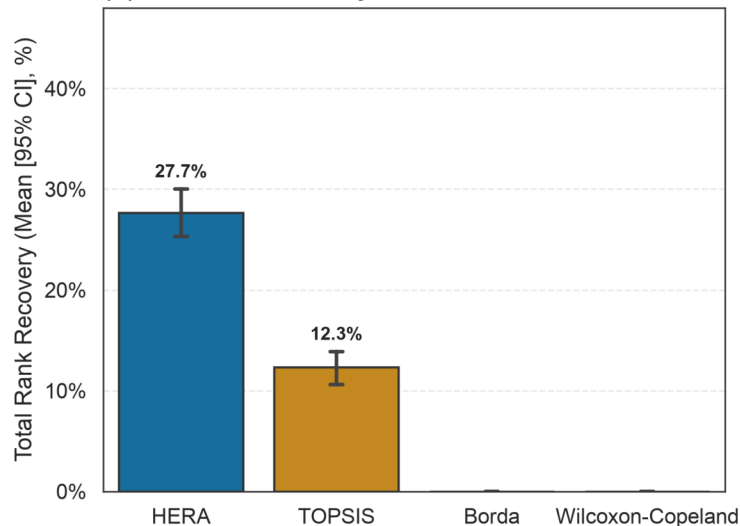
## Pooled Distribution across Core Conditions

$N \in \{10, 12, 14\}$ ,  $\sigma \in \{2\%, 4\%, 6\%, 8\%, 10\%\}$ ,  $n = 50$ ,  $\rho = 0.0$ ;  $K = 1500$  runs per method

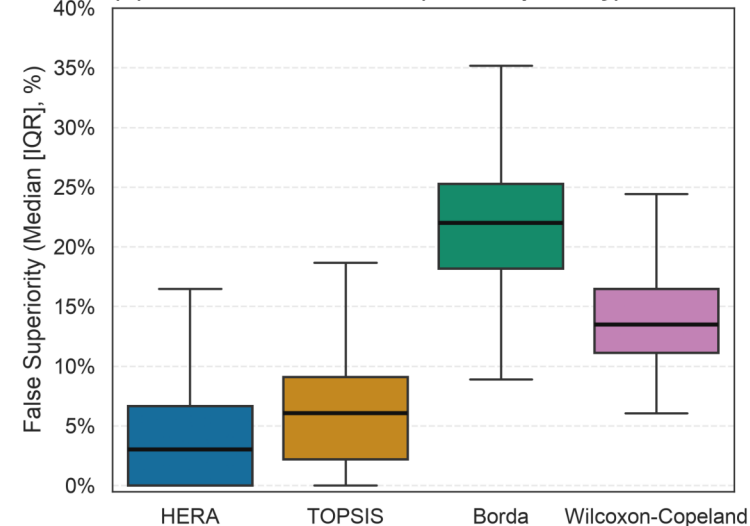
(A) Top-Choice Recovery Rate



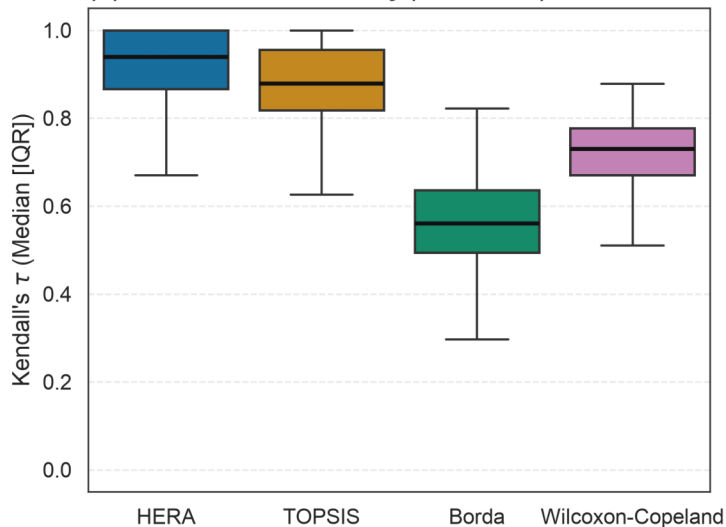
(B) Total Rank Recovery



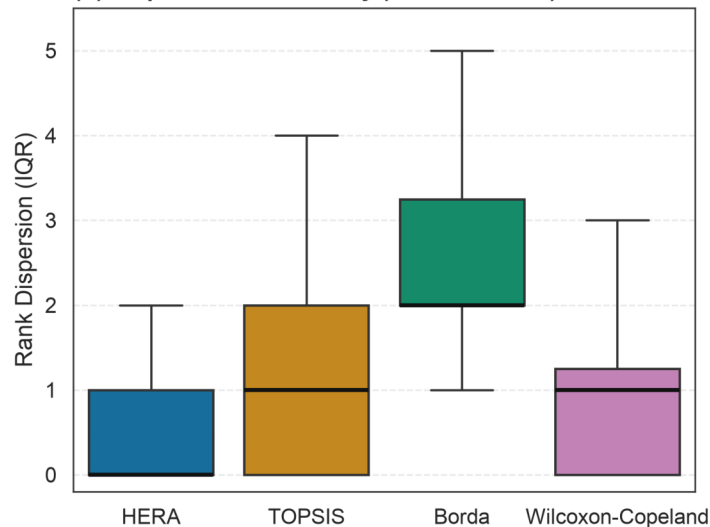
(C) Pairwise Inversion Rate (False Superiority)



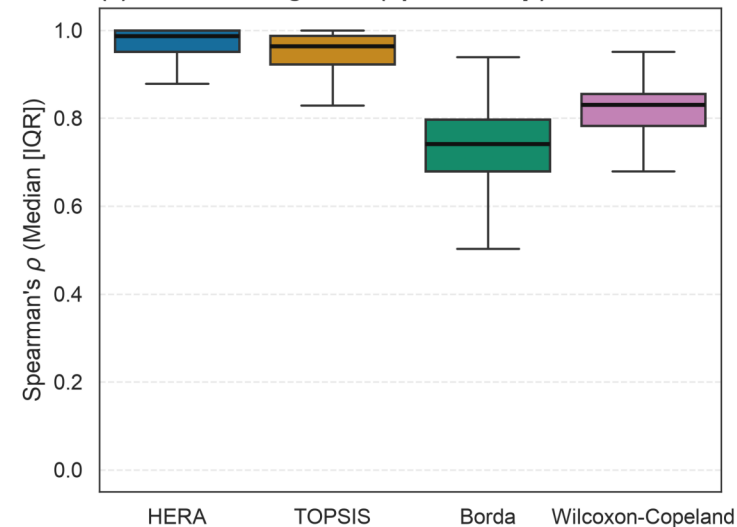
(D) Rank Correlation Fidelity (Kendall's  $\tau$ )



(E) Empirical Rank Stability (Candidate-IQR)

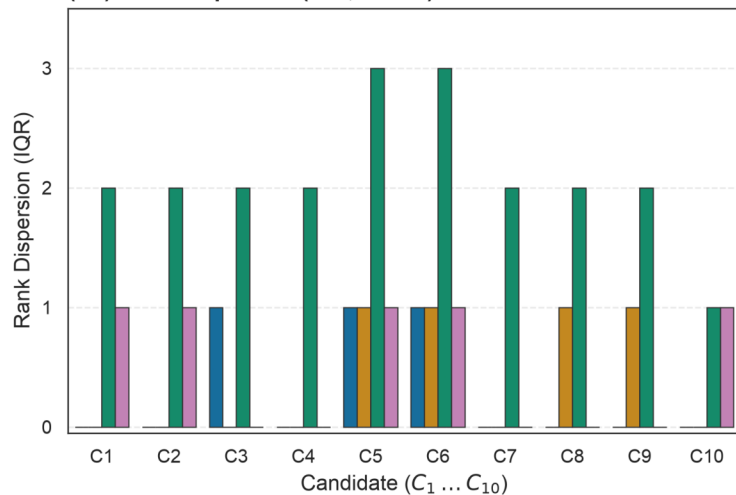


(F) Monotonic Alignment (Spearman's  $\rho$ )

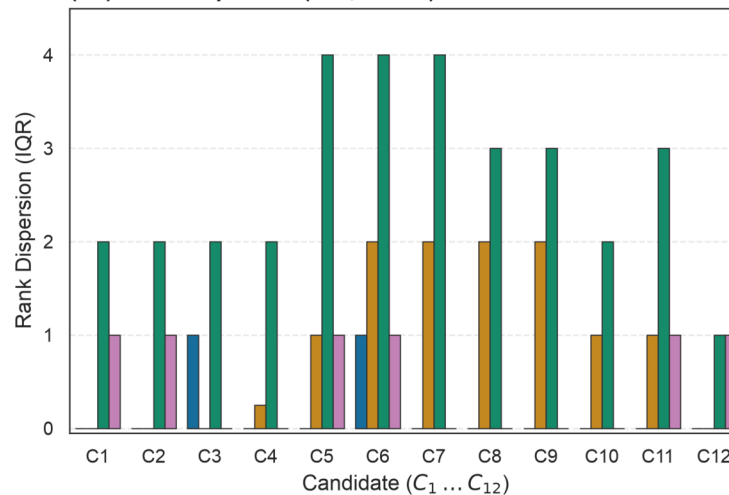


**Candidate-Level Rank Stability Across Scales**  
 $N \in \{10, 12, 14\}$ ,  $n = 50$ ,  $\sigma = 4\%$ ,  $\rho = 0.0$ ,  $K = 100$  runs per point

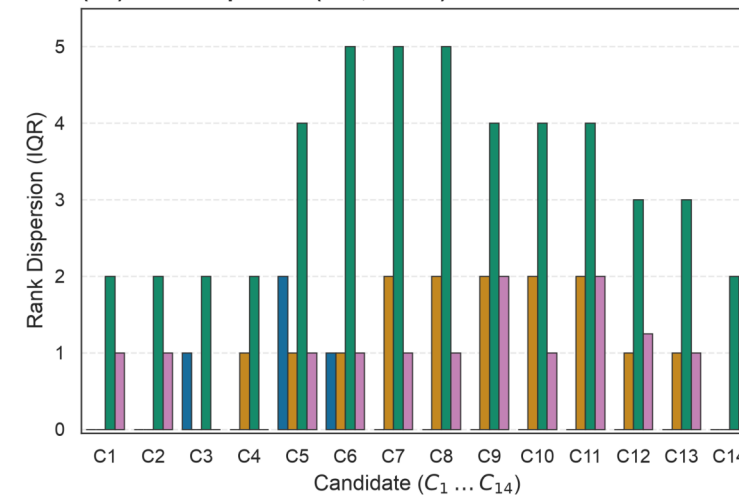
(A1) Rank Dispersion (IQR,  $N = 10$ )



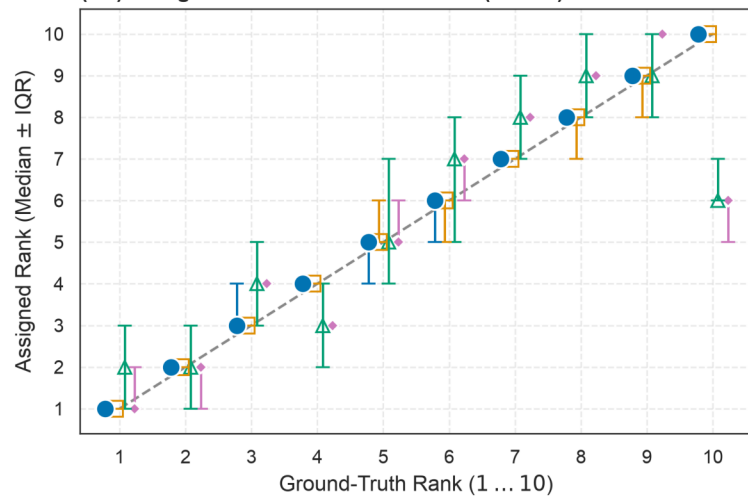
(A2) Rank Dispersion (IQR,  $N = 12$ )



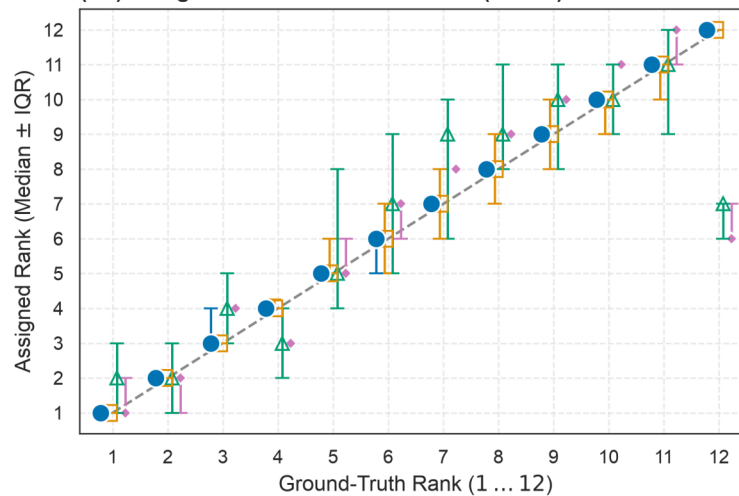
(A3) Rank Dispersion (IQR,  $N = 14$ )



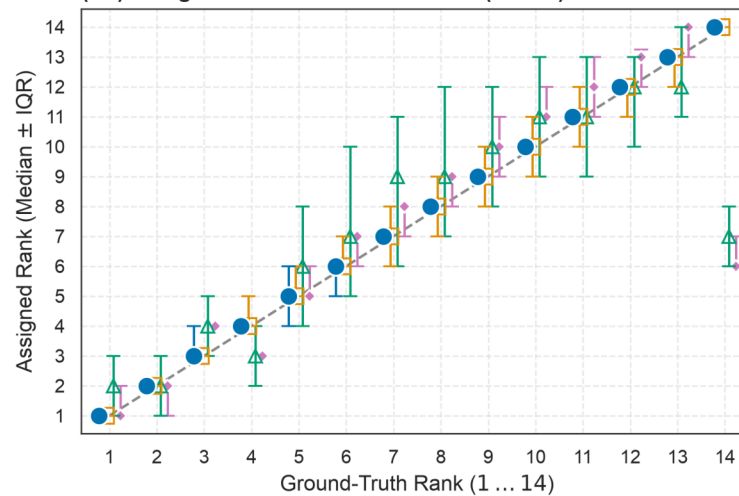
(B1) Assigned Rank vs. Ground Truth ( $N = 10$ )



(B2) Assigned Rank vs. Ground Truth ( $N = 12$ )

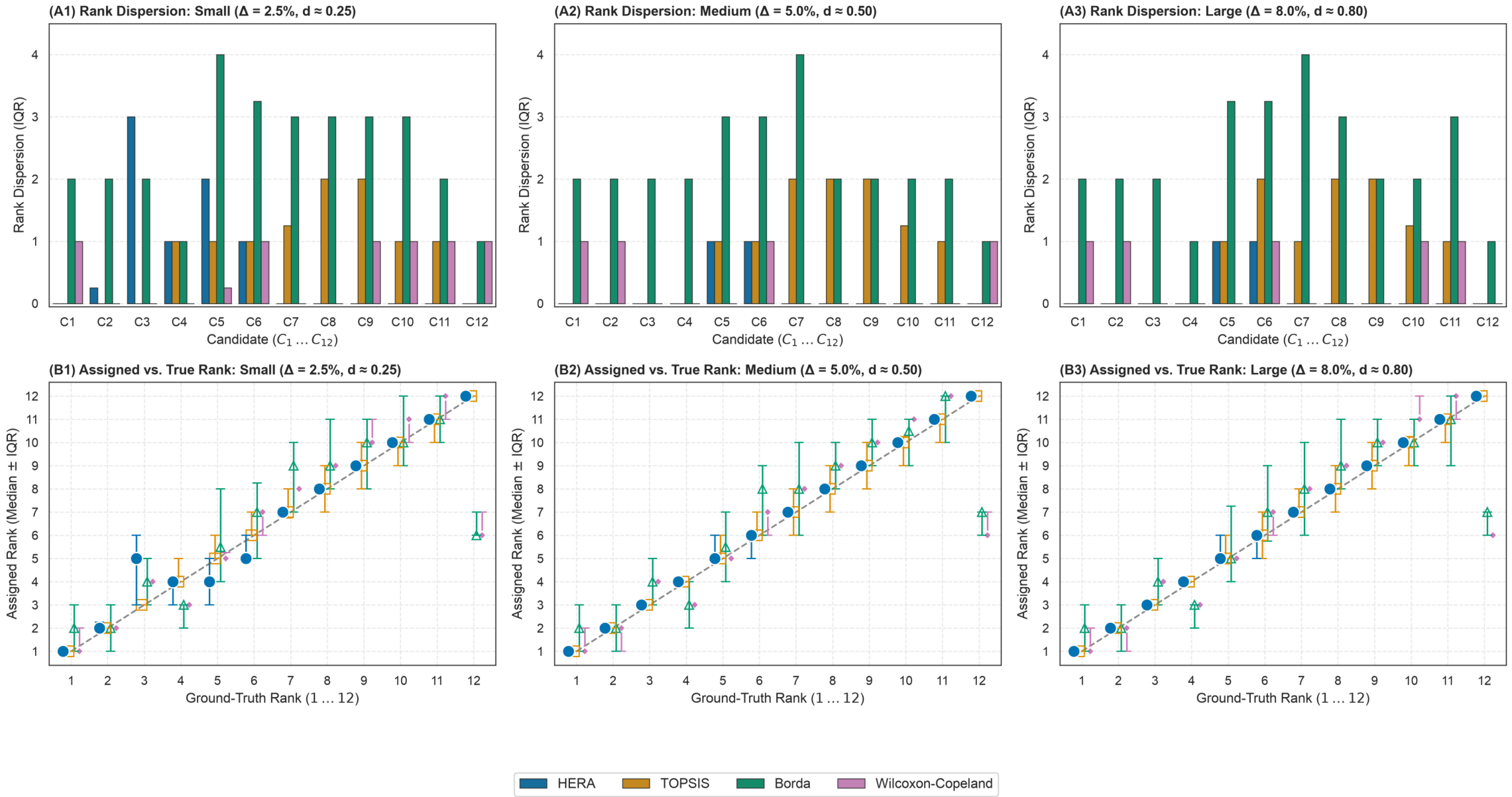


(B3) Assigned Rank vs. Ground Truth ( $N = 14$ )



Effect Size Sensitivity: Candidate-Level Rank Dispersion & Fidelity

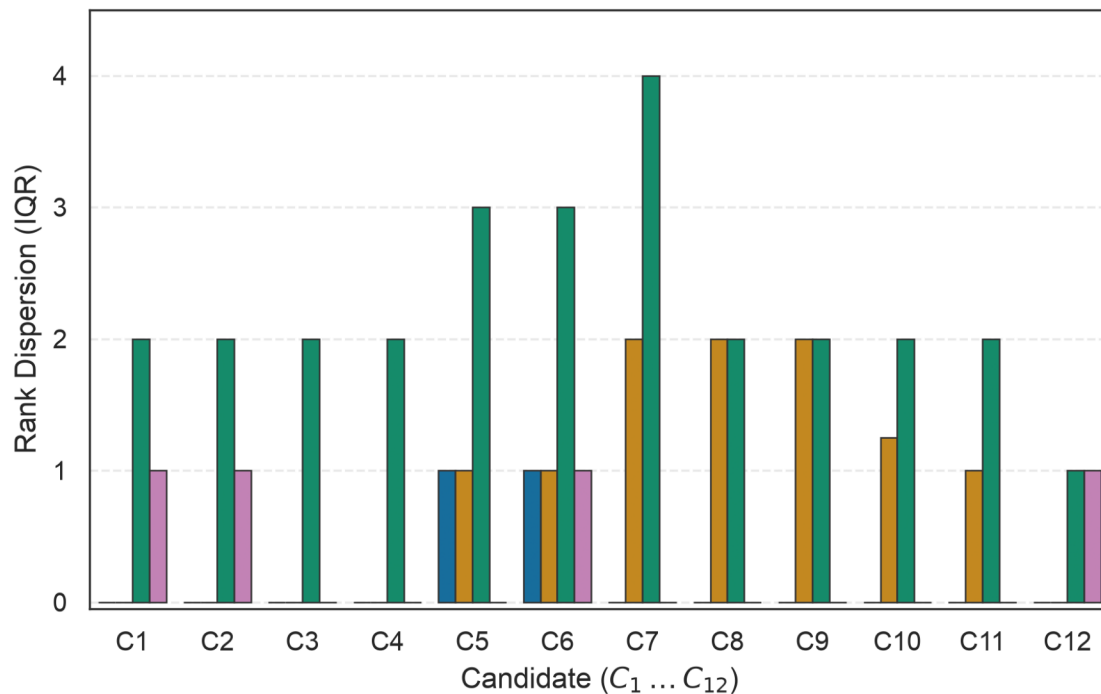
N = 12, n = 50,  $\sigma = 4\%$ ,  $\rho = 0.0$ , K = 100 runs per condition



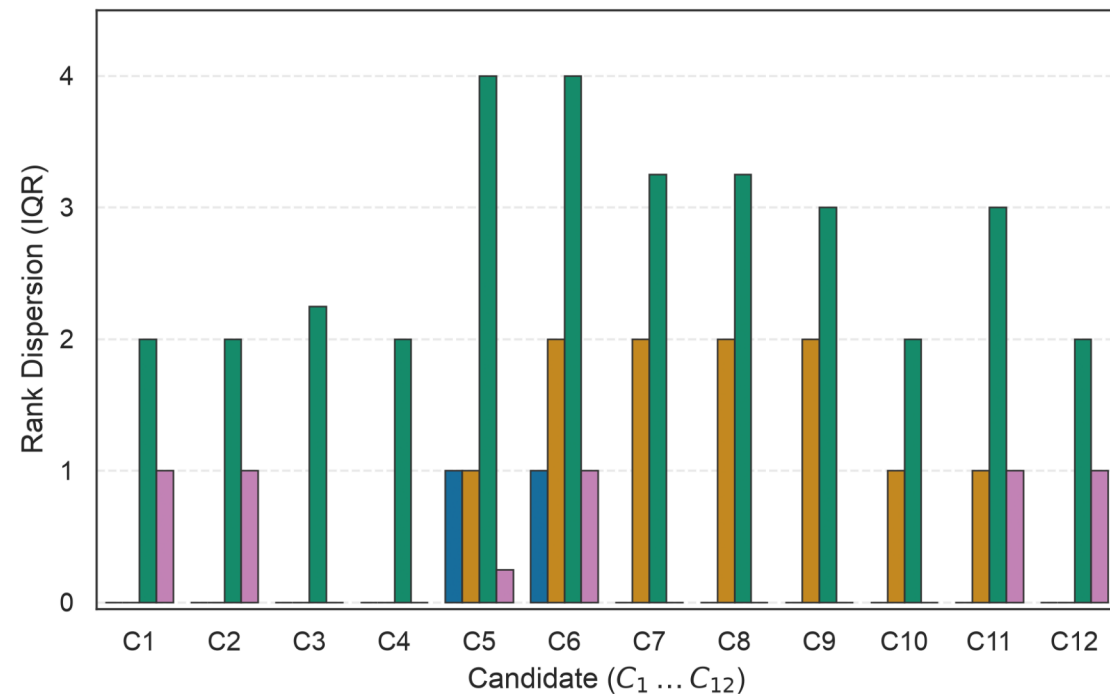
# Inter-Metric Collinearity Sensitivity: Rank Dispersion & Fidelity

$N = 12$ ,  $n = 50$ ,  $\sigma = 4\%$ , Cliff's  $d \approx 0.50$ ,  $K = 100$  runs per condition

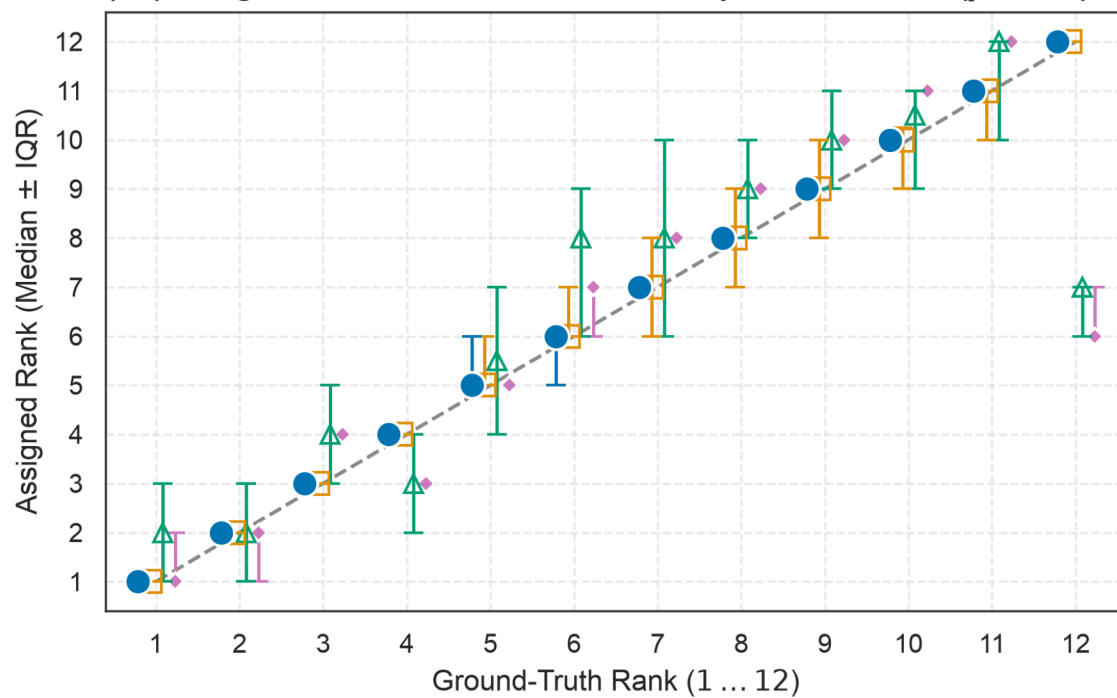
(A1) Rank Dispersion: Independent Criteria ( $\rho = 0.0$ )



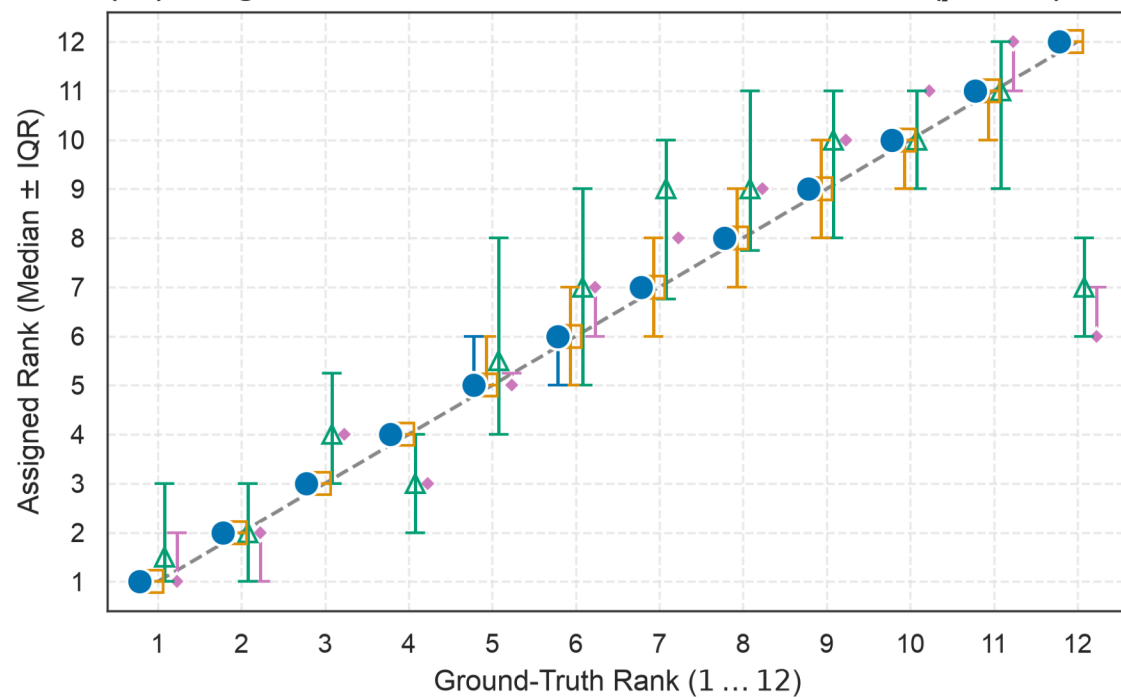
(A2) Rank Dispersion: Correlated Criteria ( $\rho = 0.5$ )



(B1) Assigned Rank vs. Ground Truth: Independent Criteria ( $\rho = 0.0$ )



(B2) Assigned Rank vs. Ground Truth: Correlated Criteria ( $\rho = 0.5$ )



HERA TOPSIS Borda Wilcoxon-Copeland